

Lead Finder Agent (Prospect Discovery System)

Project Overview

Lead Finder Agent is an AI-powered automated prospect discovery system designed to identify businesses that may need website development, redesign, or digital services.

The system automatically searches businesses from public directories and platforms, analyzes their websites, extracts contact information, and generates qualified prospect spreadsheets for outreach.

Primary Goal

Automate business lead generation for agencies, freelancers, and software companies.

Core Features

Business Discovery

Automatically find businesses from:

- Google Maps
- Justdial
- IndiaMart
- Clutch
- Company websites

Future Support:

- LinkedIn
 - Yelp
 - Yellow Pages
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Website Analysis

Detect:

- No website
- Old website
- Missing SSL
- Poor mobile responsiveness
- Missing SEO metadata
- Slow loading websites
- Poor UI/UX

Contact Extraction

Extract:

- Company Name
- Website
- Email
- Phone Number
- Social Links
- Industry

Future:

- Decision maker enrichment
 - Founder/Owner details
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Lead Qualification

Generate lead scores:

- High Potential
- Medium Potential
- Low Potential

Based on:

- Website quality
 - Missing features
 - Online presence
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Export System

Generate:

- CSV
- Excel spreadsheets

Daily automated prospect lists.

Recommended MVP Scope

Initial MVP Features

Included

- Google Maps scraping
- Website existence detection
- Basic website analysis
- Contact extraction
- MongoDB storage
- CSV export
- Basic dashboard

Excluded Initially

- LinkedIn scraping
 - Proxy rotation
 - AI outreach automation
 - Email sending
 - Cold outreach
 - n8n workflows
 - Advanced AI scoring
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Technology Stack

Frontend

- Next.js 15
 - Tailwind CSS
 - ShadCN UI
 - TanStack Table
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Backend

- Node.js
 - Express.js
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Scraping Engine

- Playwright
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Database

- MongoDB Atlas
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AI Service (Phase 2)

- FastAPI
 - Ollama
 - OpenAI API (optional)
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Export System

- ExcelJS
 - CSV Writer
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System Architecture

High-Level Flow

User Search ↓ Backend API ↓ Google Maps Scraper ↓ Business Data Extraction ↓ Website Analyzer ↓ Contact Extraction ↓ MongoDB Storage ↓ CSV/Excel Export ↓ Frontend Dashboard

Folder Structure

Frontend

/frontend |—— app |—— components |—— services |—— hooks |—— types |—— lib |——
utils

Backend

/backend |—— src | |—— scrapers | |—— services | |—— controllers | |—— routes | |——
models | |—— exports | |—— utils | |—— config

AI Service

/ai-service |—— routes |—— services |—— prompts |—— models |—— utils

Frontend Features

Search Page

Inputs:

- Business Keyword
- Location

Example:

- Gym Ahmedabad
 - Restaurant Mumbai
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Leads Dashboard

Display:

- Company Name
 - Website
 - Email
 - Phone
 - Lead Score
 - Website Status
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Export Options

Buttons:

- Export CSV
 - Export Excel
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Backend Features

Google Maps Scraper

Responsibilities

- Open Google Maps
- Search businesses
- Scroll results
- Extract:
 - Name
 - Website

- Phone
- Address
- Ratings

Technology

- Playwright
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Website Analyzer

Checks

- Website existence
 - HTTPS/SSL
 - Mobile responsiveness
 - Meta tags
 - Website title
 - Broken links
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Contact Extractor

Extract

- Emails
- Phone numbers
- Social links

Methods

- Regex extraction
 - HTML parsing
 - Contact page crawling
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Export System

Formats

- CSV
- XLSX

Libraries

- ExcelJS
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Database Design

Collection: leads

Fields:

- companyName
 - website
 - email
 - phone
 - address
 - industry
 - leadScore
 - websiteStatus
 - source
 - createdAt
-

Collection: jobs

Fields:

- keyword
 - location
 - status
 - totalLeads
 - createdAt
-

API Design

Search Leads

POST /api/leads/search

Request: { "keyword": "Gym", "location": "Ahmedabad" }

Get Leads

GET /api/leads

Export Leads

GET /api/export/csv

Website Analysis Rules

High Potential Lead

Conditions:

- No website
 - Old website
 - No SSL
 - Poor mobile experience
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Medium Potential

Conditions:

- Website exists but outdated
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Low Potential

Conditions:

- Modern optimized website
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Future Enhancements

AI Analysis

Use:

- Ollama
- OpenAI

Capabilities:

- AI lead scoring
 - AI website analysis
 - AI cold email generation
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Automation

Possible tools:

- n8n
 - BullMQ
 - Redis
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Outreach Automation

Future:

- Email sending
 - LinkedIn outreach
 - WhatsApp messaging
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MVP Development Roadmap

Phase 1

- Backend setup
 - Playwright scraper
 - MongoDB integration
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Phase 2

- Website analyzer
 - Contact extraction
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Phase 3

- CSV export
 - Dashboard frontend
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Phase 4

- AI scoring
 - Automation workflows
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Recommended Initial Workflow

Input: Keyword + Location

Process: 1. Scrape Google Maps 2. Extract business details 3. Visit websites 4. Analyze website quality 5. Save leads 6. Export spreadsheet

Output: 50–100 qualified leads.

Recommended MVP Goal

Focus only on:

- Google Maps scraping
- Website quality detection
- CSV export

Avoid:

- LinkedIn scraping
- Proxies
- Complex automation
- AI agents

until MVP is stable.

Deployment

Frontend

- Vercel

Backend

- Railway / Render / VPS

Database

- MongoDB Atlas
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Estimated MVP Timeline

Basic Working MVP

5–10 days

Production Version

1–3 months depending on scaling and automation requirements.

Final Summary

Lead Finder Agent is an automated prospect discovery platform that:

- finds businesses,
- analyzes their digital presence,
- extracts contact data,
- and generates qualified sales leads for outreach.